

Question 1: Welcome Function

Write a function named `welcome()` that prints the following message:

```
Welcome to Python Programming!
```

Call the function three times.

Question 2: Addition Function

Write a function `add(a, b)` that accepts two numbers and prints their sum.

Example:

```
Enter first number: 10
Enter second number: 20

Sum = 30
```

Question 3: Largest Number

Write a function `largest(a, b)` that accepts two numbers and prints the larger number.

Question 4: Even or Odd

Write a function `is_even(number)` that checks whether a number is even or odd.

Print an appropriate message.

Question 5: Square of a Number

Write a function `square(number)` that returns the square of a number.

Print the returned value.

Question 6: Area of a Rectangle

Write a function `area(length, width)` that returns the area of a rectangle.

Take the values from the user.

Question 7: Temperature Converter

Write a function `celsius_to_fahrenheit(celsius)` that converts Celsius to Fahrenheit.

Formula:

$$F = (C \times 9 / 5) + 32$$

Return the converted value.

Question 8: Voting Eligibility

Write a function `check_vote(age)` that returns `True` if a person is eligible to vote; otherwise, return `False`.

Display an appropriate message based on the returned value.

Question 9: Greeting Function

Write a function `greet(name)` that accepts a person's name and prints:

```
Hello, <name>! Have a great day.
```

Call the function for three different names.

Question 10: Calculator Function

Write a function `calculator(a, b, operation)` that performs one of the following operations:

- Addition
- Subtraction
- Multiplication
- Division

Use the value of `operation` to decide which calculation to perform.

Question 11: Factorial Using Functions

Write a function `factorial(number)` that returns the factorial of a number.

Example:

```
Input : 5  
Output: 120
```

Question 12: Prime Number Checker

Write a function `is_prime(number)` that returns `True` if the number is prime; otherwise, return `False`.

Use the returned value to display an appropriate message.

Question 13: Fibonacci Series

Write a function `fibonacci(n)` that prints the first `n` Fibonacci numbers.

Example:

```
Input : 7
```

```
Output:
```

```
0
1
1
2
3
5
8
```

Question 14: Power Function

Write a function `power(base, exponent)` without using Python's `**` operator.

Example:

```
power(2,5)
```

```
Output:
```

```
32
```

Question 15: Count Vowels

Write a function `count_vowels(text)` that returns the total number of vowels present in a string.

Example:

```
Input:
Programming
```

```
Output:  
3
```

Question 16: Reverse a String

Write a function `reverse_string(text)` without using slicing (`[::-1]`).

Return the reversed string.

Question 17: Function with Default Arguments

Write a function:

```
calculate_interest(principal, rate=8, years=1)
```

Calculate simple interest using:

```
SI = (Principal × Rate × Years) / 100
```

Call the function using:

- Only principal
 - Principal and rate
 - Principal, rate, and years
-

Question 18: Variable-Length Arguments

Write a function `average(*numbers)` that accepts any number of numeric arguments and returns their average.

Example:

```
average(10,20,30)
```

```
Output:  
20
```

Question 19: Recursive Sum

Write a recursive function `sum_n(n)` that returns the sum of the first `n` natural numbers.

Example:

Input:
5

Output:
15

Question 20: Menu-Driven Calculator

Create a menu-driven program using functions.

Menu:

1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit

Requirements:

- Create a separate function for each mathematical operation.
- Continuously display the menu until the user selects **Exit**.
- Validate division by zero.